IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

OPTIMISATION

Thursday, 30 April 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

This is an OPEN BOOK examination.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) :

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OPTIMISATION

1. a) Consider the (continuous) function

$$f(x_1,x_2) = x_1^2 + x_2^2 + \cos(x_1^2).$$

Show that the function is not convex, that is that the Hessian matrix is not positive definite for all (x_1,x_2). Show that the function has infinitely many stationary points and that the point $(0,0)$ is a local strict minimizer.

(Note that $(0,0)$ is a global minimizer and all other stationary points are saddle points – but you do not have to show this.)

[4 marks]

b) Consider the (continuous) function

$$f(x_1,x_2) = 2x_1^2 - x_1x_2 + x_2^2 - 3x_1 + e^{2x_1+x_2}$$

and the point $x_0 = (0,0)$. Determine a descent direction d for f at x_0. Let $x_N = x_0 + \alpha d$. Determine a value of α such that $f(x_N) < f(x_0)$. [4 marks]

c) Consider the optimization problem

$$\begin{aligned} &\min_{x_1,x_2} x_1, \\ &(x_1-1)^2 + (x_2-1)^2 \leq 1, \\ &(x_1-1)^2 + (x_2+1)^2 \leq 1. \end{aligned}$$

i) Plot on the (x_1,x_2) plane the level lines of the objective function and the admissible set. Show that the admissible set does not have any regular point. [4 marks]

ii) Find the optimal solution of the considered problem and show that such a solution does not satisfy the necessary conditions of optimality. Explain why this is the case. [4 marks]

d) Consider the optimization problem

$$\begin{aligned} &\min_{x_1,x_2} x_1^2 - x_2^2, \\ &x_1 - x_2 = 1, \\ &x_1 \geq 0, \qquad x_2 \geq 0. \end{aligned}$$

Eliminate the equality constraint and determine an optimal solution for the problem. Show that this is a strict minimizer. [4 marks]

2. Consider the constrained optimization problem

$$\min_{x_1,\, x_2} \overbrace{(x_1-4)^2+(x_2-6)^2}^{f(x_1,x_2)}$$
$$x_1^2 \le x_2,$$
$$x_2 \le 4.$$

a) Plot on the (x_1,x_2) plane the level lines of the objective function and the admissible set. [4 marks]

b) Show that all points in the admissible set are regular points and solve, graphically, the considered problem. [4 marks]

c) Write the first order necessary conditions of optimality and determine the candidate optimal solution $x_\star$. [8 marks]

d)

Sufficient condition of optimality (for convex functions over a convex set)
A candidate optimal point $x_\star$ is a (global) minimizer if

$$\nabla f^\top(x_\star)(x-x_\star) \ge 0$$

for all x in the admissible set.

Use the sufficient condition in the box (note that the function f is convex and the admissible set is convex) to show that the candidate solution $x_\star$ determined in part c) is a minimizer. [4 marks]

3. Consider the optimization problem

$$\min_{x_1,\, x_2} \; -x_1 x_2,$$
$$x_1 + 2x_2 - 4 = 0.$$

a) By plotting the admissible set and the level lines of the objective function determine the optimal solution $P_\star$ of the problem. [4 marks]

b) By using first order necessary conditions of optimality and second order sufficient conditions of optimality determine the optimal solution (again $P_\star$) of the considered problem. [4 marks]

c) Write the sequential penalty function for the considered problem and determine its stationary point. Show that, as the penalty parameter ε tends to 0, the stationary point of the sequential penalty function tends to $P_\star$. [4 marks]

d) Write the exact penalty function for the problem and show that its stationary point coincides with $P_\star$. [8 marks]

4. a) Consider the function

$$f(x) = \begin{cases} (x-1)^2, & \text{if } x \leq -1, \\ (x+1)^2, & \text{if } x \geq 1, \\ \psi(x) & \text{if } x \in [-1,1], \end{cases}$$

where ψ is any function such that f is smooth (and convex). Show that Newton's iteration with initial point $x_0 = 1$ does not convergence and yields the sequence $\{x_0, x_1, x_2, \cdots\} = \{1, -1, 1, \cdots\}$. [4 marks]

b) Consider the function

$$f(a,b,c) = a^2b + b^2c + c^2a.$$

i) By using first order necessary conditions of optimality determine the unique stationary point P of the function f .

(Hint: since you are told that the function has only one stationary point, identify such a point from inspection, that is do not explicitly solve the equation $\nabla f = 0$.)

Show that the Hessian matrix of f at the stationary point P is the zero matrix (that is all the entries of the Hessian matrix at the stationary point are equal to 0). [4 marks]

ii) To classify the stationary point determined in part b.i) show that arbitrarily close to P there are points P_1 and P_2 such that

$$f(P_1) > f(P) = 0 > f(P_2),$$

hence conclude that P is a saddle point. [4 marks]

c) Consider the function

$$f(x) = 9x - 4\ln(x-7),$$

for $x \geq 7$.

i) Show that the function is convex and radially unbounded, hence compute its global minimizer. [4 marks]

ii) Write explicitly Newton's iteration for the minimization of the function f. Hence run three steps of Newton's iteration with initial values $x_0 = 7.5$ and $x_0 = 8$. Explain the behaviour of the obtained results. [4 marks]